Predicting Coastal Flooding Events

XGBoost & Physics-Aware Feature Engineering

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Model Selection: XGBoost Classifier

- Baseline: XGBoost Regressor
  - Prove the effective to tabular time-series data
  - Robustness & interpretability of tree-base model
- Regressor -> Classifier
  - The objective is "flooding or not": classification task

Feature Engineering 1: Physical Context

Understand the physical relationship between "water level" and "flooding"

- Difference from the threshold
  - How close we are to flooding

$$DailyMax - (Mean + 1.5 \times Std)$$

- Rate of change of water level
  - Capture sudden surge events

$$SeaLevel_t - SeaLevel_{t-1}$$

Feature Engineering 2: Seasonality & Trend

Capture seasonality & trend pattern

- Periodic encoding

$$\sin\left(\frac{2\pi \times Month}{12}\right), \cos\left(\frac{2\pi \times Month}{12}\right)$$

- Year trend $(Year - 1950)/70$

Feature Engineering 3: Inertia & Volatility

Flooding is a continuous event; it does **not occur and end randomly**

- Flood Inertia
    - The sum of flood labels (0/1) over the past 7 days

- Volatility
    - The standard deviation of water levels over the past 7 days

Imbalance Strategy

- Leaderboard: high F1 scores (0.94), but MCC is almost zero
  - Speculated: test set has high flooding rate (90% up?)
  - Distribution Shift (flooding rate of training set ~= 68%)

- Training
  - Set “scale_pos_weight”

- Submit: active thresholding
  - Best threshold for local validation: 0.46 (maximizing MCC)
  - Final submission threshold: 0.15 (maximizing Recall)

Conclusion

- Model selection: XGBoost Classifier
- Feature Engineering: physical properties, trends, and inertia
- Strategic Tuning: Decision thresholds are flexibly adjusted based on the race metric (F1) and data distribution characteristics.

Thanks for attention

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